

Mini Project 1.1: Link Budget Simulation — Console Output

Output of: python link_budget.py

=== BASELINE ===

700 MHz (low-band)	f= 700 MHz PL@0.5km= 83.3 dB PRX= -31.3 dBm margin= 68.7 dB max range= 1358.01 km
28 GHz (mmWave)	f= 28000 MHz PL@0.5km= 115.4 dB PRX= -63.4 dBm margin= 36.6 dB max range= 33.95 km

Range ratio (700 MHz / 28 GHz) = 40.0x (= 28000/700 = 40x in frequency)

=== TASK 1: 28 GHz with 24 dBi TX gain ===

28 GHz, 24 dBi	f= 28000 MHz PL@0.5km= 115.4 dB PRX= -54.4 dBm margin= 45.6 dB max range= 95.68 km
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Range recovered: 33.95 -> 95.68 km (x2.82); +9 dB gives $10^{(9/20)} = 2.82x$

=== TASK 2: add 3.5 GHz mid-band ===

3.5 GHz (mid-band)	f= 3500 MHz PL@0.5km= 97.3 dB PRX= -45.3 dBm margin= 54.7 dB max range= 271.60 km
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=== TASK 3: extra 8 dB shadowing/fading margin ===

700 MHz :	1358.01 km ->	540.63 km (shrinks to 40% of original)
3.5 GHz :	271.60 km ->	108.13 km (shrinks to 40% of original)
28 GHz :	33.95 km ->	13.52 km (shrinks to 40% of original)

=== TASK 4: SNR and capacity (B = 10 MHz, noise = -104 dBm) ===

700 MHz (low-band)	d= 0.1 km: PRX= -17.3 dBm, SNR= 86.7 dB, C= 287.9 Mbps
700 MHz (low-band)	d= 1 km: PRX= -37.3 dBm, SNR= 66.7 dB, C= 221.4 Mbps
700 MHz (low-band)	d= 10 km: PRX= -57.3 dBm, SNR= 46.7 dB, C= 155.0 Mbps
28 GHz (mmWave)	d= 0.1 km: PRX= -49.4 dBm, SNR= 54.6 dB, C= 181.4 Mbps
28 GHz (mmWave)	d= 1 km: PRX= -69.4 dBm, SNR= 34.6 dB, C= 115.0 Mbps
28 GHz (mmWave)	d= 10 km: PRX= -89.4 dBm, SNR= 14.6 dB, C= 49.0 Mbps

=== TASK 5: double TX power vs double bandwidth ===

700 MHz	d= 0.5 km: base= 241.4 2x power= 251.4 (x1.04) 2x bandwidth= 462.9 (x1.92)
28 GHz	d= 0.5 km: base= 135.0 2x power= 145.0 (x1.07) 2x bandwidth= 250.0 (x1.85)
28 GHz	d= 10 km: base= 49.0 2x power= 58.8 (x1.20) 2x bandwidth= 79.0 (x1.61)